



SIMATS ENGINEERING

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Course Name: CLOUD COMPUTING AND BIG DATA ANALYTICS

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1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.

Answer:

Cloud computing is the delivery of computing resources such as servers, storage, databases, networking, software, and processing power over the Internet. Users can access these resources whenever required without owning and maintaining the physical infrastructure themselves.

The following characteristics make cloud computing different from traditional on-premise computing:

1. On-Demand Self-Service

Users can automatically obtain computing resources such as servers, storage, and applications whenever they need them without requiring direct interaction with the service provider.

2. Broad Network Access

Cloud services are available over the Internet and can be accessed from different devices such as laptops, smartphones, tablets, and desktop computers.

3. Resource Pooling

The cloud provider pools its computing resources and dynamically allocates them to multiple users according to their requirements. Virtualization technology helps in efficiently sharing these resources.

4. Rapid Elasticity

Cloud resources can be quickly increased or decreased based on demand. For example, an online shopping website can use additional servers during heavy traffic.

5. Measured Service

Cloud usage is monitored and measured. Users generally pay according to the resources they consume, such as

storage, processing power, or data transfer.

Difference from Traditional On-Premise Computing

In traditional on-premise computing, an organization purchases, owns, and maintains its own hardware and software infrastructure. In cloud computing, these resources are provided by a cloud service provider through the Internet. Cloud computing also provides better scalability, flexibility, and a pay-as-you-use pricing model.

2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.

Answer:

Cloud computing evolved through several stages of development in hardware, networking, virtualization, and Internet-based software.

1. Mainframe Computing

In the early stage, powerful mainframe computers were used by multiple users through terminals. This introduced the concept of sharing computing resources among many users.

2. Personal Computers

The development of personal computers provided individual users with their own processing power and storage. However, organizations had to purchase and maintain separate hardware for every user.

3. Computer Networks

The development of computer networks allowed computers to communicate and share resources. Local Area Networks (LANs) and Wide Area Networks (WANs) enabled file sharing, printer sharing, and communication between systems.

4. Internet Evolution

The growth of the Internet allowed users to access information and applications from anywhere in the world. Web-based software reduced the need to install applications locally.

5. Virtualization

Virtualization allowed a single physical server to run multiple virtual machines. This improved hardware utilization, reduced costs, and enabled the flexible allocation of computing resources.

6. Distributed Computing and Large Data Centers

Cloud providers developed large data centers containing thousands of servers, storage systems, and networking

equipment. These resources could be managed and shared across different locations.

7. Modern Cloud Platforms

The combination of high-speed Internet, virtualization, distributed computing, automation, and large-scale data centers resulted in modern cloud computing platforms.

Evolution Flow:

Mainframe Computing → Personal Computers → Computer Networks → Internet → Virtualization → Distributed Computing → Large Data Centers → Cloud Computing

These technological developments enabled cloud platforms to provide scalable, flexible, reliable, and on-demand computing services over the Internet.

3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.

Answer:

Server virtualization is the process of dividing a single physical server into multiple independent virtual servers using a software layer called a hypervisor. Each virtual machine can run its own operating system and applications.

Role of Server Virtualization in Cloud Computing

1. Efficient Resource Utilization

Virtualization allows multiple virtual machines to share the resources of a single physical server, thereby reducing unused capacity.

2. Resource Sharing

The physical resources such as CPU, memory, and storage can be allocated to multiple users.

3. Scalability

Virtual machines can be created, modified, or removed quickly according to user requirements.

4. Cost Reduction

Organizations can reduce hardware and maintenance costs by running multiple virtual servers on fewer physical machines.

5. Isolation

Each virtual machine operates independently. Problems in one virtual machine generally do not directly affect other

virtual machines.

Virtualization as an Enabling Technology

Virtualization is a technology that abstracts physical hardware and creates virtual resources such as virtual machines, virtual storage, and virtual networks.

Cloud Computing as a Service Model

Cloud computing is a service delivery model that provides computing resources such as infrastructure, platforms, and software to users over the Internet.

Difference Between Virtualization and Cloud Computing

Virtualization	Cloud Computing
It is an enabling technology	It is a service delivery model
Creates virtual resources from physical hardware	Delivers computing resources as services
Mainly focuses on hardware abstraction	Focuses on providing services to users
Can exist without cloud computing	Often uses virtualization technology
Example: VMware virtual machines	Example: Microsoft Azure cloud services

Therefore, virtualization provides the technical foundation for efficient resource management, while cloud computing uses virtualization along with networking, automation, storage, and data centers to deliver IT resources as services.

4. Discuss the role of data centers in delivering cloud services.

Answer:

A data center is a specialized facility that contains servers, storage systems, networking equipment, power systems, cooling systems, and security infrastructure. Data centers form the physical foundation of cloud computing.

Role of Data Centers in Delivering Cloud Services

1. Providing Computing Resources

Data centers contain powerful servers that process user requests and run cloud applications, websites, databases, and other services.

2. Providing Storage

Data centers provide large-scale storage for storing files, databases, application data, backups, images, and videos.

3. Providing Network Connectivity

High-speed network connections connect data centers to the Internet. This allows users to access cloud services from different locations.

4. Supporting Scalability

Cloud providers can add more servers, storage, and networking resources when the number of users or workload increases.

5. Ensuring Availability and Reliability

Data centers use redundant servers, backup power supplies, multiple network connections, and backup systems to ensure that cloud services remain available.

6. Providing Security

Data centers protect cloud infrastructure using physical security, firewalls, access control, encryption, monitoring, and other security mechanisms.

7. Supporting Backup and Disaster Recovery

Cloud providers maintain copies of data and applications in multiple locations. If one data center fails, another data center can continue providing the service.

Cloud Service Delivery Process:

User Device → Internet → Cloud Network → Data Center → Servers and Storage → Cloud Service

Therefore, data centers are essential for delivering cloud services because they provide the physical infrastructure required for computing, storage, networking, security, scalability, reliability, and disaster recovery.

5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

Answer:

IaaS, PaaS, SaaS, and XaaS are different cloud service models that provide different levels of computing resources and services to users.

Service Model	Full Form	Services Provided	User Mainly Manages	Real-World Example
IaaS	Infrastructure as a Service	Virtual machines, servers, storage, and networks	Operating system, applications, and data	Microsoft Azure Virtual Machines
PaaS	Platform as a Service	Development platform and runtime environment	Application code and data	Google App Engine
SaaS	Software as a Service	Complete software applications	User data and application settings	Google Maps
XaaS	Everything as a Service	Any IT resource delivered as a service	Depends on the service	Dropbox

IaaS – Infrastructure as a Service

IaaS provides basic computing infrastructure such as virtual machines, storage, servers, and networks through the Internet.

Example: Microsoft Azure Virtual Machines.

The cloud provider manages the physical hardware, while the user manages the operating system, applications, and data.

PaaS – Platform as a Service

PaaS provides a ready-to-use platform for developing, testing, and deploying applications.

Example: Google App Engine.

The cloud provider manages the servers, operating system, and runtime environment, allowing developers to focus mainly on application development.

SaaS – Software as a Service

SaaS provides complete software applications over the Internet. Users can access the software without installing or maintaining the underlying infrastructure.

Example: Google Maps.

The service provider manages the servers, operating system, application, and infrastructure.

XaaS – Everything as a Service

XaaS is a general term used to describe the delivery of almost any IT resource or service through the Internet.

Examples include:

- Storage as a Service
- Database as a Service
- Security as a Service
- Artificial Intelligence as a Service
- Network as a Service

Example: Dropbox provides cloud storage as a service.